Literature status of Scheepers' conjecture

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Lane 3, `agent_lane_03`, model GPT5.6

11 August 2026

Status. Literature already answered: Peng gives a counterexample under CH. This is a retrieval and identification note, not a new proof.

Source question

Tsaban and Zdomskyy [1], Section 4.4 on PDF page 13, define a *wQN space* and state Scheepers' conjecture as

$$X \text{ is wQN} \iff X \models \mathrm{S}_1(\Gamma, \Gamma).$$

They then ask whether this conjecture is provable. Their Corollary 24(3) proves that the conjecture holds in the Laver model.

The same section records the exact bridge needed below. In the proof of Corollary 24(1), $C_p(X)$ is an α_2 space exactly when $X \models \mathrm{S}_1(\mathcal{C}_\Gamma, \mathcal{C}_\Gamma)$; the remark following Corollary 24 states that X is wQN exactly when $X \models \mathrm{S}_1(\mathcal{C}_\Gamma, \mathcal{C}_\Gamma)$.

Later answer

Peng [2] explicitly announces in the title and abstract that Scheepers' conjecture is refuted under CH. The abstract-level main theorem is: assuming CH, there is a subset X of the reals such that $C_p(X)$ has property α_2, but X does not satisfy $\mathrm{S}_1(\Gamma, \Gamma)$.

By the equivalences quoted in the source paper, the first property says that X is wQN. Peng's X therefore witnesses

$$X \text{ is wQN}, \qquad X \not\models \mathrm{S}_1(\Gamma, \Gamma),$$

which is exactly a counterexample to the nontrivial direction of the source conjecture. The provenance is explicit: Peng says the construction refutes Scheepers' conjecture, so this belongs in `literature_already_answered` rather than `literature_implied_answers`.

Scope and conclusion

The result is a consistent counterexample, not an unconditional ZFC example. Assuming the usual consistency background, it nevertheless answers the source's precise provability question negatively. Combined with the Laver-model result in the source, it gives independence of the conjecture from ZFC.

A bounded search through 11 August 2026 checked the exact title, DOI, Crossref metadata, the author's institution announcement, and later papers citing Peng's resolution. The publisher PDF is not open access and is therefore not copied into this packet; the DOI and official metadata remain stable evidence.

References

[1] B. Tsaban and L. Zdomskyy, *Hereditarily Hurewicz spaces and Arhangel'skii sheaf amalgamations*, arXiv:1004.0211v2 (2012).

[2] Y. Peng, *Scheepers' conjecture and the Scheepers diagram*, Trans. Amer. Math. Soc. **376** (2023), 1199–1229, doi:10.1090/tran/8787.